2025 AMC 10A Problem 1 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 1. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 1

Problem:

Andy and Betsy both live in Mathville. Andy leaves Mathville on his bicycle at $1{:}30$, traveling due north at a steady 8 miles per hour. Betsy leaves on her bicycle from the same point at $2{:}30$, traveling due east at a steady 12 miles per hour. At what time will they be exactly the same distance from their common starting point?

Answer Choices:

A. $3{:}30$

B. $3{:}45$

C. $4{:}00$

D. $4{:}15$

E. $4{:}30$

Solution:

Let t be the number of hours after $1{:}30$. Then, if we represent Andy's distance as d_A and Betsy's distance as d_B, we get

$$d_A = 8t \qquad d_B = 12(t-1)$$

Thus, for these to be equal, we set $8t = 12(t-1) = 12t - 12$, so $4t = 12$, or $t = 3$. This means that they reach this equidistant point 3 hours after $1{:}30$, or at $\textbf{(E) 4:30}$.